

Nirbhay Singh

Technical Lead-RADAR System(AESA), Mixed-Signal & Digital Beamforming



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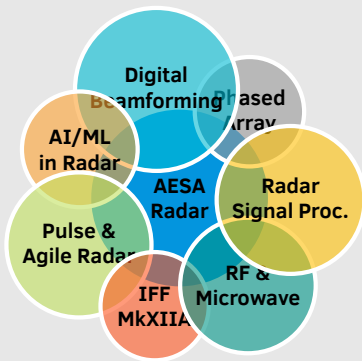
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/Aboutmywork/

Technical Skills

Overview



Programming

Embedded C • JESD204 • SerDes

Python • Deep Learning • ML for Radar

MATLAB • Simulink • DSP

VHDL • FPGA • RFSoc

Education

M.Tech., Communication & Signal Processing

Indian Institute of Technology (IIT) Madras
CGPA: 9.32 / 10

B.Tech., Electronics & Communication Engineering

Bundelkhand Institute of Engineering & Technology
80% Graduated with Honours

About Me

Senior FPGA and AESA Radar Systems Engineer with 10+ years of experience designing and delivering defence-grade Active Electronically Scanned Array (AESA) radar systems with large-scale multi-channel architectures (160×16 channels).

Expert in direct RF sampling systems, deterministic multi-chip/multi-board synchronization using JESD204B/C, and adaptive digital beamforming (DBF) using both phase-based (PB) and amplitude/phase (PA) techniques.

Strong background in FPGA-based real-time DSP, ultra-low phase noise system design, and AI-driven radar signal processing for target detection and classification.

Proven experience across end-to-end radar system design including RF, mixed-signal hardware, FPGA firmware, and system-level integration achieving MDS levels up to -156 dBm.

Experience -- 10+ Years

Nov 2023 – **Deputy Manager – Radar & RF Transceiver Systems (Team Lead)**
Bharat Electronics Limited (BEL)

Present

- Technical Lead for design and realization of large-scale AESA radar systems with **160×16 coherent channels** across multi-chip and multi-board architectures
- Architected **direct RF sampling-based synchronization system** using **JESD204B/C**, enabling deterministic latency alignment across distributed radar nodes
- Designed and implemented **adaptive digital beamforming (DBF)** using **phase-based (PB) and phase-amplitude (PA) techniques** for high-resolution beam steering
- Achieved **receiver sensitivity (MDS) of -156 dBm** through optimized RF front-end, clocking and low-noise design
- Developed **FPGA-based real-time DSP chains** including:
 - Digital Down Conversion (DDC) & Digital Pulse Compression
 - Doppler processing (FFT-based) & CFAR-based target detection
- Designed **multi-channel waveform generation engines** supporting pulse, FMCW, chirp and frequency-agile radar modes
- Implemented high-speed **RF ADC/DAC interfaces up to 12 GHz sampling** with JESD204 deterministic synchronization
- Led **multi-board clock distribution and synchronization architecture** ensuring phase coherence across large antenna arrays
- Applied **AI/ML techniques for radar target detection and classification**, including neural network-based models for clutter rejection and object identification
- Designed and validated **ultra-low phase noise clocking systems** critical for Doppler and micro-Doppler accuracy
- Led system integration, calibration, and field validation of AESA radar subsystems with cross-functional teams

2017 – 2022

Senior Radar & Digital Systems Engineer Bharat Electronics Limited (BEL)

- Designed and integrated radar receiver and exciter subsystems for pulse and phased-array radar platforms
- Developed coherent radar processing chains including pulse compression, MTI and Doppler FFT processing
- Performed radar performance analysis: range equation, SNR, dynamic range and detection probability
- Implemented and validated IFF Mk-XIIA compliant digital transceiver modules
- Conducted phase noise and jitter analysis impacting Doppler and micro-Doppler extraction
- Supported EMI/EMC mitigation in RF and digital radar subsystems
- Conducted extensive laboratory testing and field validation

2016

- Performed complete **hardware design cycle**: schematic, PCB layout, signal integrity (SI), power integrity (PI), and high-speed RF design
- Digital subsystem design, modeling and integration
- Receiver chain analysis including gain, noise figure, phase noise and sensitivity estimation
- Radar link budget and signal processing gain calculations
- Supported system-level validation of radar transceiver modules

Research

2022 – 2024 **Postgraduate Research – Radar AI** Indian Institute of Technology Madras**Thesis:** Target Detection and Classification Using Deep Learning for Surveillance Radar

- Deep learning models for radar target detection and classification
- Analysis of phase noise and micro-Doppler effects on ML-based radar systems
- Tools: MATLAB, Python, Deep Learning Toolboxes

Publications

- **Nirbhay Kumar Singh**, “Advanced Deep Learning Technique for Digital Predistortion to Improve Power Amplifier Linearity”. IEEE MAPCON2024
- **Nirbhay Kumar Singh**, Malwinder Singh, Shaik Abdul Subhan, Seema, “Phase Noise Effects on Micro-Doppler Extraction for Radar Target Classification Using Machine Learning”. IEEE APMC2024, Bali, Indonesia
- **Nirbhay Kumar Singh**, “Multi-board Synchronization of n-channel Direct RF Digital Transceiver for Active Phased Array Radar using JESD204 Interface”. IEEE I2CT · Apr 6, 2024
- **Nirbhay Kumar Singh**, Vikas Kumar, T Venakatamuni, “Ultra-Low Phase Noise, Digital Transceiver for High Range Resolution Pulsed Agile Radar”. 8th IEEE International Radio Conference, Mauritius · May 1, 2023
- **Nirbhay Kumar Singh**, T Venakatamuni, “A Wide-band SWaP System for VSWR and Power Measurement with Temperature Compensation Technique”. IEEE MAPCON2022 · Dec 14, 2022
- **Nirbhay Kumar Singh**, Vikas Kumar, Fouziya c, T Venakatamuni, A Programmable, Multimode Operational 3U-VPX Based Digital Transceiver & Processing Module For CIT-MKXIIA IFF
- **Nirbhay Kumar Singh**, T Venakatamuni, “A method to Enhance Power Amplifier Linearity using Time delay estimation & correction Technique”, IEEE, 2022 International Conference for Advancement in Technology (ICONAT) Goa, India. Jan 21-22, 2022
- Vikas Kumar, **Nirbhay Kumar Singh**, T Venakatamuni, “HIGH-RESOLUTION FMCW GENERATION FOR MICRO DOPPLER IN DRONE DETECTION RADAR”, International Radar Symposium India(IRSI) 2019 9. **Nirbhay Kumar Singh**, Vikas Kumar, Fouziya c, T Venakatamuni, “Performance Analysis of High-Resolution Radar Waveforms for Naval Tracking Radar”. EWCI 2018
- **Nirbhay Kumar Singh**, Fouziya C, Venkatamuni, “Radar Target Emulator For Naval Tracking Radar Using FPGA”, URSI-RCRS 2017

Patents

- A method and system of synchronization of an n-channel direct RF digital transceiver with high-speed serial JESD204 protocol. Patent no: 202341083757
- A Method to generate a wideband exciter signal having multi-pulse-coded waveforms with test-target simulation in a radar receiver. Patent no: 202341004980

- Wide-band SWAP System and Method for VSWR and Power Measurement using Digital Signal Processing with Temperature Compensation. Patent no: 202241070066
- Compact and Programmable Digital Transceiver for Software-defined Radar. Patent no: 202041013933
- System and Method to Convert Non-Coherent FMCW Radar to Coherent Radar. Patent no: 202141009908
- A SYSTEM AND METHOD FOR REALIZING ULTRA-LOW PHASE NOISE, FAST SETTLING, COMPACT PHASE COHERENT AND ENERGY EFFICIENT DIGITAL SOURCE GENERATOR. Patent no: 202341030479

Awards

- R&D Excellence Award (2020–2023)
- Excellent R&D Award – Innovation Category
- Best Technical Poster Awards (Defence AI, Unmanned Systems)

Additional Information

- Languages: English (Professional), Hindi (Native)
- Open for relocation; Skilled Worker Visa sponsorship required
- 10+ years experience in Defence & Aerospace R&D